

UNIT 3: ENVIRONMENTAL BIOLOGY

Complete Study Notes — Part 2 of 2 (Topics 6–10)

Biodiversity · Conservation · Genetics · Evolution · Applied Ecology

Biodiversity & Hotspots

Conservation Biology

Population Genetics

Evolution & Applied

PART 2 OVERVIEW — TOPICS 6–10

Topic	Sub-topics covered in Part 2	Exam weight
6. Biodiversity	Types (alpha/beta/gamma), indices, hotspots (global+India), HIPPO threats, endemic species, mass extinctions	VERY HIGH
7. Conservation Biology	In-situ/ex-situ methods, protected areas, IUCN Red List, biosphere reserves, Indian wildlife laws, CBD/CITES/Ramsar	VERY HIGH
8. Population Genetics	Hardy-Weinberg, genetic drift, founder effect, bottleneck, inbreeding depression, N_e	HIGH
9. Evolution	Natural selection types, speciation (allopatric/sympatric), coevolution, adaptive radiation	HIGH
10. Applied Ecology	Island biogeography, metapopulation, ecological footprint, restoration ecology	HIGH

TOPIC 6: BIODIVERSITY — COMPLETE COVERAGE

6.1 Levels of Biodiversity

Level	Definition	Measured by	Examples
Genetic diversity	Variety of genes and genotypes within a species or population	Allelic richness, heterozygosity, DNA sequence variation	Different rice varieties; wolf population genetics
Species diversity	Variety of species in a community	Species richness (S), Shannon H', Simpson's D, Pielou's J (evenness)	1,000 species in tropical forest; 10 in arctic tundra
Ecosystem diversity	Variety of ecosystems, habitats, and ecological communities in a region	Number and type of ecosystems; landscape indices	India: deserts + rainforests + mangroves + alpine meadows
Functional diversity	Variety of functional traits and ecological roles	Functional richness, functional evenness, functional divergence	Diversity of pollinator types, seed dispersers, decomposers

6.2 Biodiversity Indices — All Formulae with Interpretation

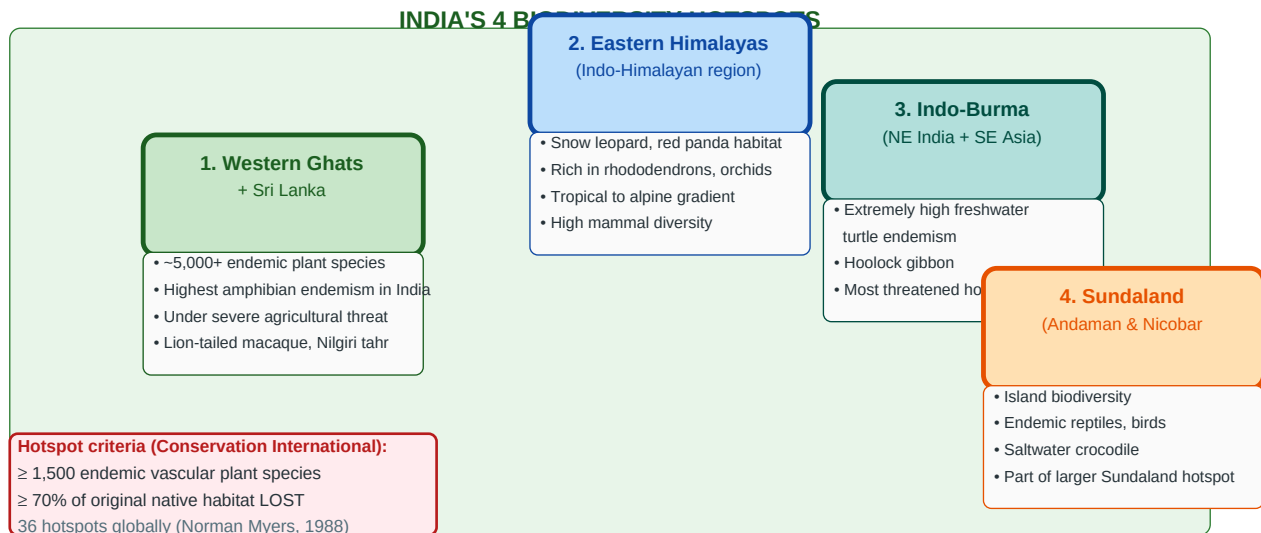
Index	Formula	Range	Interpretation	Exam frequency
Species Richness (S)	S = total number of species	1 to ∞	Higher = more species; simplest measure; ignores abundance	HIGH
Shannon-Wiener (H')	$H' = -\sum(\pi_i \times \ln \pi_i)$	0 to $\ln(S)$	$H'=0$: one species; accounts for richness AND evenness; most widely used	VERY HIGH
Simpson's Index (D)	$D = \sum \pi_i^2$	0 to 1	D near 0 = high diversity; D near 1 = low diversity (one dominates)	HIGH
1 – D (Simpson's diversity)	$1 - \sum \pi_i^2$	0 to 1	Reverse of D; higher = more diverse	MEDIUM
Pielou's Evenness (J)	$J = H' / \ln(S)$	0 to 1	$J=1$: perfectly even; $J=0$: one species dominates	HIGH
Margalef Richness (DMg)	$DMg = (S-1) / \ln(N)$	—	Accounts for sample size; older index	MEDIUM
Whittaker's β	$\beta = \gamma/\alpha$	≥ 1	$\beta=1$: identical communities; higher = more turnover	HIGH

Shannon-Wiener Quick Calculation Method

Step 1: Count total individuals (N). Step 2: Find proportion of each species ($\pi_i = n_i/N$). Step 3: Calculate $\pi_i \times \ln(\pi_i)$ for each species. Step 4: Sum all values and negate. H' is MAXIMISED when all species have equal abundance (perfect evenness). $H' = 0$ when only ONE species present (no diversity at all).

6.3 Global Biodiversity Hotspots

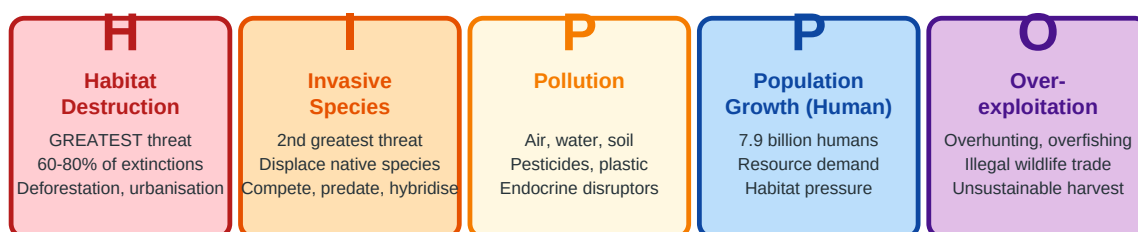
Biodiversity Hotspot (Norman Myers, 1988, refined by Conservation International): Area with (1) $\geq 1,500$ endemic vascular plant species AND (2) has lost $\geq 70\%$ of its original native vegetation. Currently **36 hotspots globally**, covering only $\sim 2.5\%$ of Earth's land surface but containing $>50\%$ of plant endemics and $\sim 42\%$ of terrestrial vertebrate endemics.



India's Hotspot	States/Region	Key endemic species	Major threats
Western Ghats + Sri Lanka	Kerala, Karnataka, Tamil Nadu, Maharashtra, Goa, Sri Lanka	Lion-tailed macaque, Nilgiri tahr, Malabar giant squirrel; >5,000 plant species; highest amphibian endemism	Plantation agriculture (tea, coffee, cardamom), urbanisation, quarrying
Eastern Himalayas	Arunachal Pradesh, Sikkim, Meghalaya, Assam, Bhutan, Nepal	Red panda, snow leopard, clouded leopard, golden langur; rich orchids and rhododendrons	Hydroelectric dams, jhum cultivation, climate change affecting altitudinal zones
Indo-Burma	Nagaland, Manipur, Mizoram, Tripura, Assam; Myanmar, Thailand, Cambodia	Hoolock gibbon, Eld's deer, Irrawaddy dolphin; extremely high freshwater turtle diversity	Slash-and-burn, illegal wildlife trade, dam construction
Sundaland	Andaman & Nicobar Islands (India portion)	Andaman serpent eagle, Nicobar megapode, saltwater crocodile; endemic island species	Tsunami impact, invasive species, tourism pressure, resource extraction

6.4 HIPPO — Five Threats to Biodiversity

H-I-P-P-O — Five Major Threats to Biodiversity (E.O. Wilson)



Threat	Details	Global examples	Indian examples
Habitat Destruction (H)	GREATEST cause of biodiversity loss worldwide; 60-80% of documented extinctions linked to habitat loss	Amazon deforestation, Indonesian peat forest clearing for palm oil, wetland drainage	Conversion of Western Ghats forests to tea/coffee; mangrove loss along coasts
Invasive Species (I)	Second greatest threat; exotic species outcompete, predate, hybridise with, or bring disease to native species	Nile perch in Lake Victoria eliminated 200+ cichlid species; brown tree snake in Guam	Water hyacinth (Eichhornia) choking water bodies; Lantana camara in forest undergrowth

Threat	Details	Global examples	Indian examples
Pollution (P)	Chemical pollution (pesticides, heavy metals), plastic (microplastics in food chains), light/noise pollution	DDT thinning raptor eggshells; endocrine disruptors in aquatic systems; ocean plastic	Pesticide runoff in farmland; industrial effluent in rivers killing fish
Population growth — Human (P)	8 billion humans create massive demand for food, water, energy, space; indirect driver of all other threats	Global resource consumption exceeding Earth's biocapacity since 1970s	India's 1.4 billion population creating intense pressure on forests, rivers, coasts
Overexploitation (O)	Unsustainable hunting, fishing, collection exceeding population's ability to reproduce	Atlantic cod collapse; shark finning; bushmeat hunting in Africa	Tiger poaching; sea turtle egg collection; overfishing in coastal waters

6.5 Mass Extinctions & Current Biodiversity Crisis

Mass Extinction	When	Cause	% species lost
1st — Ordovician-Silurian	~443 mya	Glaciation, sea level fall	~86%
2nd — Late Devonian	~375 mya	Multiple causes; anoxic oceans	~75%
3rd — Permian-Triassic (Great Dying)	~252 mya	Volcanic eruptions (Siberian Traps), climate change	~96% — largest ever
4th — Triassic-Jurassic	~201 mya	Volcanic eruptions (CAMP), climate change	~80%
5th — Cretaceous-Paleogene (K-Pg)	~66 mya	Asteroid impact + volcanism; killed dinosaurs	~76%
6th — CURRENT (Anthropocene)	Ongoing since ~1500 CE	Human activities (HIPPO); rate 1,000× background	10,000+ species/year risk extinction

6.6 Important International Days & Events

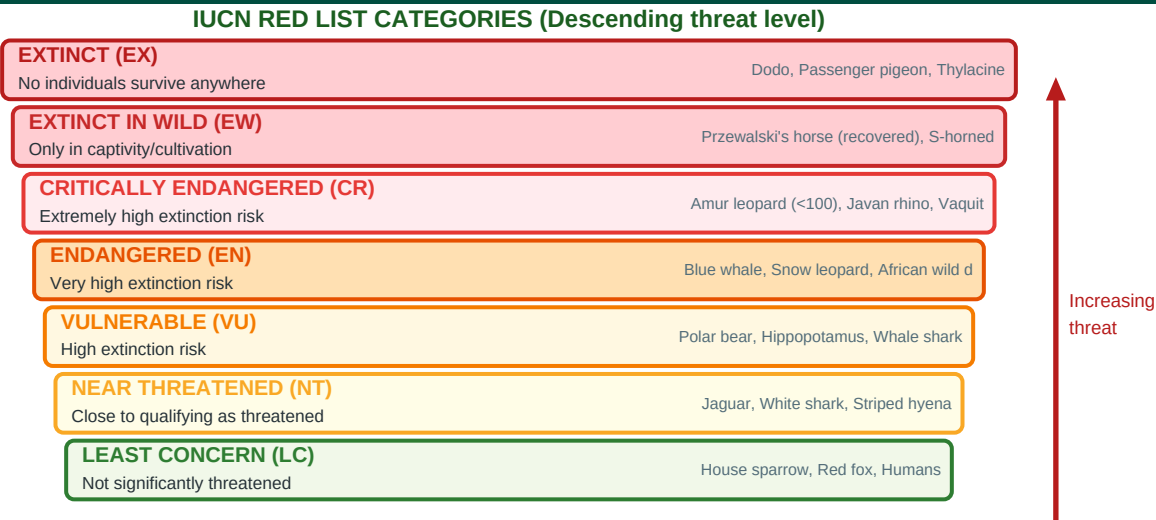
Event/Day	Date	Significance
World Environment Day	June 5	Declared at Stockholm Conference 1972; annual theme set by UNEP
International Day for Biological Diversity	May 22	Commemorates adoption of CBD text (1992)
World Wildlife Day	March 3	Commemorates signing of CITES (1973)
World Wetlands Day	February 2	Commemorates signing of Ramsar Convention (1971)
Earth Day	April 22	Since 1970; largest environmental civic movement
COP (Conference of Parties) to CBD	Every 2 years	COP15 = Kunming-Montreal 2022 (30×30 target)

TOPIC 7: CONSERVATION BIOLOGY — PROTECTED AREAS & LAWS

7.1 In-situ vs Ex-situ Conservation — Complete Comparison

Feature	In-situ Conservation	Ex-situ Conservation
Meaning	Conservation IN natural habitat	Conservation OUTSIDE natural habitat
Definition	Protecting species within their natural ecosystem	Maintaining species in controlled settings away from nature
Methods	National parks, wildlife sanctuaries, biosphere reserves, conservation reserves, community reserves, sacred groves, Ramsar wetlands	Zoological parks (zoos), botanical gardens, aquaria, seed banks, gene banks, cryopreservation, captive breeding, tissue culture
Advantages	Natural behaviour maintained; natural selection continues; ecosystem processes intact; self-sustaining; cost-effective	Can save species from imminent extinction; controlled breeding; genetic management; disease protection; ex-situ research
Disadvantages	Species still face threats (poaching, habitat degradation); hard to control; difficult for rare species	High cost; artificial environment; loss of natural behaviours; limited space; animals may not survive reintroduction
Indian examples	Jim Corbett NP, Kaziranga WS, Sundarbans BR, Chilika Ramsar site, Sacred groves (Devvans)	Zoological parks: Delhi, Chennai, Kolkata; NBPGR New Delhi (seed bank); Nandankanan (white tiger captive breeding)

7.2 IUCN Red List — All 9 Categories



Category	Code	Criteria (simplified)	Examples
Extinct	EX	No reasonable doubt that last individual has died; exhaustive surveys found none	Dodo (1681), Passenger pigeon (1914), Thylacine (1936), Great auk (1844)
Extinct in Wild	EW	Survives only in cultivation, captivity, or as naturalised population outside native range	Arabian oryx (re-introduced), Scimitar-horned oryx, Wyoming toad
Critically Endangered	CR	≥50% population decline in 10 yrs OR ≥80% decline in 3 generations OR <250 mature individuals	Amur leopard (<100), Javan rhino (<75), Vaquita porpoise (<10!), Saola

Category	Code	Criteria (simplified)	Examples
Endangered	EN	≥50% decline in 3 generations OR <2,500 mature individuals OR very restricted range	Blue whale, Snow leopard, African wild dog, Irrawaddy dolphin, Gharial
Vulnerable	VU	≥30% decline OR <10,000 mature individuals OR restricted but not critical range	Polar bear, Hippopotamus, Whale shark, Indian elephant, Gaur
Near Threatened	NT	Close to qualifying as VU; may do so in near future	Jaguar, Great white shark, Striped hyena, Indian grey wolf
Least Concern	LC	Not currently qualifying for higher categories	House sparrow, Red fox, Nile crocodile, Snow goose, Humans
Data Deficient	DD	Inadequate information to assess extinction risk	Many invertebrates, deep-sea species, rare plants
Not Evaluated	NE	Has not yet been assessed against Red List criteria	Most of Earth's described species

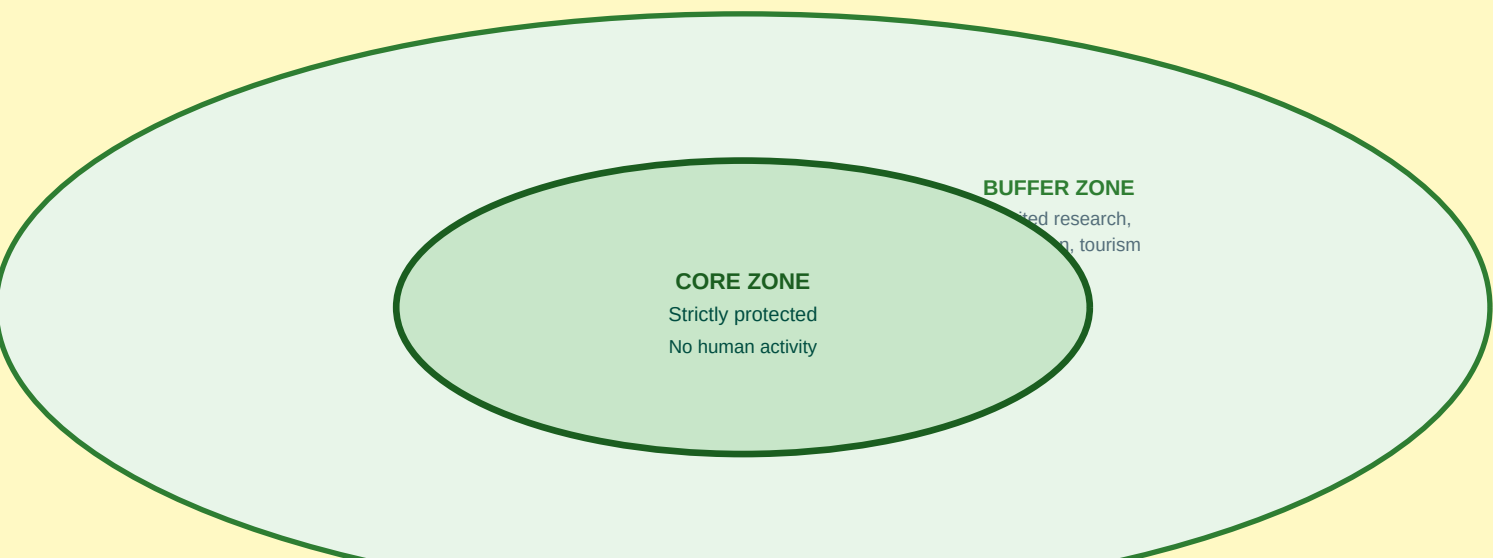
IUCN Memory Trick

EX-EW-CR-EN-VU-NT-LC-DD-NE 'Every Elephant Can Eat Very Nice Leaves Daily Now' THREATENED = CR + EN + VU (these three together = 'threatened' in legal/policy contexts)

7.3 Protected Areas in India

Category	Number in India	Legal basis	Key features
National Parks	106	Wildlife Protection Act 1972, Sec 35	Highest protection; no human activity inside; no buffer; boundaries fixed by law
Wildlife Sanctuaries	567	WPA 1972, Sec 26A	Some human activities permitted; less strict than NP; private land may exist inside
Biosphere Reserves	18 (12 UNESCO)	UNESCO MAB Programme	Three zones: core, buffer, transition; largest; multiple use; human habitation in transition
Tiger Reserves	54	Project Tiger 1973; WPA 1972	Critical Tiger Habitat (core) + buffer; no human activity in core
Conservation Reserves	100+	WPA 1972, Sec 36A	Community-owned lands; highest stakeholder involvement; newer category (2002 amendment)
Community Reserves	50+	WPA 1972, Sec 36C	Village commons/community land; local community manages; minimal government control
Ramsar Sites	75+	Ramsar Convention 1971	Wetlands of international importance; no strict activity prohibition but wise use principle
World Heritage Sites — Natural	7	UNESCO World Heritage Convention 1972	Outstanding universal value; Kaziranga, Sundarbans, Western Ghats, Manas, etc.

7.4 Biosphere Reserve Zonation (MAB Programme)



Zone	Activities allowed	Purpose	Example (Nilgiri BR)
Core Zone	NO human activity; strictly protected; only non-invasive research permitted	Protect biodiversity and natural processes; benchmark for environmental monitoring	Mudumalai, Nagarhole, Bandipur NPs — core
Buffer Zone	Limited research, environmental education, non-consumptive tourism permitted; no resource extraction	Protect core from human impact; research and monitoring zone	Around national parks — limited entry
Transition Zone (Cooperation Zone)	Human settlements, traditional agriculture, eco-tourism, sustainable resource use, local industries	Demonstrate sustainable development; support local livelihoods	Villages, towns, tea estates

7.5 Key Indian Environmental Laws — Conservation Focused

Act/Law	Year	Key provisions	Exam tip
Wildlife Protection Act	1972	Protects listed species (Schedules I–VI); establishes national parks and sanctuaries; bans hunting; establishes Wildlife Crime Control Bureau	Schedule I = highest protection (tiger, lion, rhino). Year 1972 always asked.
Forest Conservation Act	1980	Prior approval from Central Govt needed for any diversion of forest land to non-forest use	Controls deforestation; important for forest conservation
Biological Diversity Act	2002	Establishes National Biodiversity Authority (NBA); regulates access to biological resources; prevents biopiracy; Access and Benefit Sharing (ABS)	NBA headquarters: Chennai. Year 2002 always asked.
Environment Protection Act	1986	Umbrella legislation; gives powers to Central Govt to take measures for environmental protection	Enacted after Bhopal gas tragedy (1984). Year 1986 key.
Project Tiger	1973	World's largest tiger conservation programme; currently 54 reserves; established Critical Tiger Habitats	1 April 1973 launch; PM Indira Gandhi; 54 reserves as of 2023.
Project Elephant	1992	Elephant conservation; elephant corridors; 33 elephant reserves in India	Launched 1992; protects elephant corridors between forests

7.6 Key International Conventions — Conservation

Convention	Year	Focus	Key details
CITES	1963 (force 1975)	International trade in endangered species	3 appendices: App I = no commercial trade; App II = regulated; App III = country request. 183+ parties.
Ramsar Convention	1971 (force 1975)	Wetlands of international importance	'Wise use' principle; 2,400+ Ramsar sites globally; India has 75+ including Chilika (first, 1981), Loktak, Wular, Sambhar

Convention	Year	Focus	Key details
Convention on Biological Diversity (CBD)	1992 (Rio)	3 objectives: conservation, sustainable use, fair benefit sharing	Nagoya Protocol (2010) = ABS; Cartagena Protocol (2000) = biosafety; COP15 (2022) = Kunming-Montreal 30×30 target
Nagoya Protocol	2010 (force 2014)	Access and Benefit Sharing (ABS) of genetic resources	Prevents biopiracy; fair sharing of benefits from use of genetic resources with source countries
World Heritage Convention	1972	Sites of outstanding universal value	UNESCO administered; India has 42 sites (7 natural); Kaziranga, Sundarbans, Western Ghats, Manas, Nanda Devi, Great Himalayan NP, Khangchendzonga
Minamata Convention	2013	Mercury pollution	Named after Minamata disease (mercury poisoning in Japan)

EXAM TIPS — Conservation Biology

1. In-situ = IN nature (national parks). Ex-situ = EXITED (zoos, seed banks). Most important distinction. 2. Biosphere reserve zones: CORE (strictest) → BUFFER → TRANSITION. Core = no humans. 3. IUCN CR criteria: <250 mature individuals OR ≥80% decline in 3 generations. 4. CITES Appendix I = complete ban on commercial trade. Appendix II = regulated. 5. Ramsar = WETLANDS. Chilika Lake = India's FIRST Ramsar site (1981). 6. Project Tiger = 1973, 54 reserves. Wildlife Protection Act = 1972, Schedule I highest. 7. Biological Diversity Act = 2002, NBA = Chennai. 8. CBD adopted at Rio Earth Summit 1992. Nagoya Protocol 2010 = ABS framework.

TOPIC 8: POPULATION GENETICS — EVOLUTION IN POPULATIONS

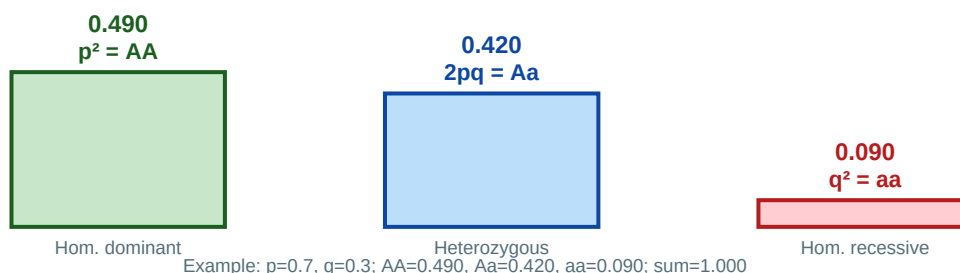
8.1 Hardy-Weinberg Equilibrium (1908)

Hardy-Weinberg Law (Godfrey Hardy & Wilhelm Weinberg, 1908): In a large, randomly mating population, allele and genotype frequencies remain CONSTANT from generation to generation in the absence of evolutionary forces. This is the null hypothesis of population genetics — deviations indicate evolution is occurring.

$$\text{HARDY-WEINBERG EQUILIBRIUM} \text{ — } p^2 + 2pq + q^2 = 1$$

Allele frequencies: p (dominant A) + q (recessive a) = 1

$$p^2 + 2pq + q^2 = 1 \text{ (genotype frequencies)}$$



Hardy-Weinberg Conditions	What would violate it (= causing evolution)
1. Large population size (no genetic drift)	Small population → Genetic drift changes frequencies randomly
2. Random mating (no sexual selection, no assortative mating)	Assortative mating → Changes genotype but not allele frequencies
3. No mutation	Mutation → Creates new alleles; changes p and q
4. No natural selection	Selection favours/disfavours certain genotypes → Changes frequencies
5. No gene flow (closed population)	Immigration/emigration → Introduces or removes alleles

H-W Formula	Variables	Application
$p + q = 1$	p = frequency of dominant allele (A); q = frequency of recessive allele (a)	Find missing allele frequency: if $p=0.7$, $q=0.3$
$p^2 + 2pq + q^2 = 1$	p^2 = AA frequency; $2pq$ = Aa frequency; q^2 = aa frequency	Calculate expected genotype frequencies
$q = \sqrt{\text{aa frequency}}$	Frequency of recessive homozygote = q^2	Find allele frequencies from phenotype data; q^2 is the only directly observable genotype for recessive trait

Hardy-Weinberg — Worked Exam Example

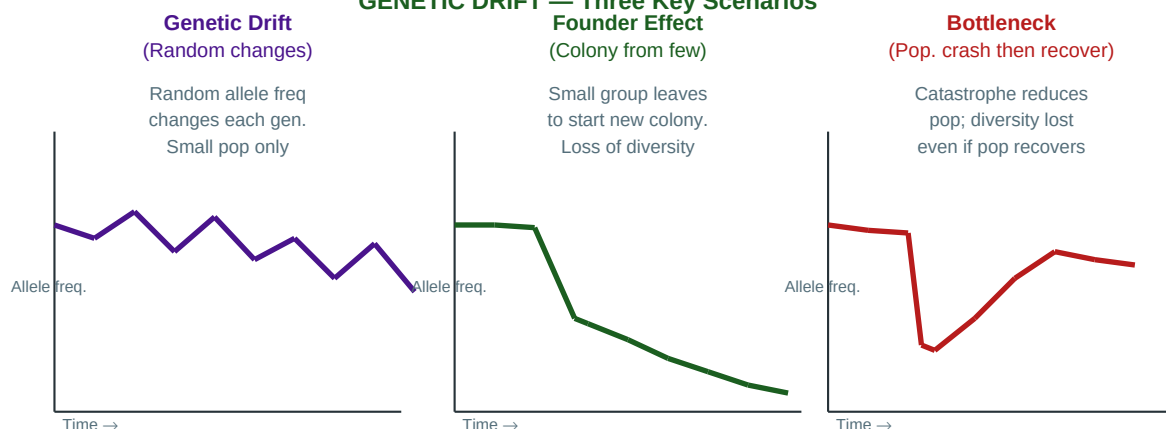
Problem: 1 in 10,000 people have albinism (aa). Find: p , q , and frequency of carriers (Aa). Step 1: $q^2 = 1/10,000 = 0.0001$ → $q = \sqrt{0.0001} = 0.01$ Step 2: $p = 1 - q = 1 - 0.01 = 0.99$ Step 3: Carriers (Aa) = $2pq = 2 \times 0.99 \times 0.01 = 0.0198 \approx 2\%$ of population are carriers Key insight: Many more people CARRY recessive allele than EXPRESS it — 198 carriers for every 1 affected person.

8.2 Mechanisms of Evolution (Microevolution)

Mechanism	Definition	Effect on genetic diversity	Example
Natural selection	Differential survival and reproduction based on phenotype fitness	Can increase or decrease diversity depending on type	Antibiotic resistance; industrial melanism in peppered moth
Genetic drift	Random changes in allele frequencies due to chance — significant in SMALL populations	Decreases diversity; can fix or eliminate alleles randomly	Amish community disorders; cheetah low diversity
Mutation	Random changes in DNA sequence — ultimate source of new alleles	Increases diversity (adds new alleles)	Sickle cell mutation in humans; antibiotic resistance genes
Gene flow (migration)	Movement of alleles between populations via migrating individuals	Homogenises populations; increases local diversity	Bird migration introducing new alleles; seed dispersal
Non-random mating	Mate choice based on phenotype (assortative mating or sexual selection)	Changes genotype but NOT allele frequencies (no H-W violation of allele freq)	Peacock tail; colour-based mate choice

8.3 Genetic Drift — Special Cases

GENETIC DRIFT — Three Key Scenarios



Special case	Definition	Key outcome	Conservation relevance
Founder effect	Small group of individuals establishes new isolated population with only subset of original alleles	Low genetic diversity; rare alleles in founder group become common or lost	Invasive species often have founder effect; island colonisation
Bottleneck effect	Population drastically reduced by disaster; survivors' alleles unrepresentative of original	Major loss of genetic diversity even if population recovers numerically	Cheetahs (past bottleneck); Northern elephant seal
Inbreeding depression	Mating between relatives increases homozygosity; exposes deleterious recessive alleles	Reduced fitness, fertility, survival in inbred individuals	Critically endangered species in small populations; Florida panther

8.4 Effective Population Size (N_e)

Concept	Definition/Formula	Significance
Effective population size (N_e)	Theoretical size of ideal random-mating population with same rate of genetic drift as actual population; always $\leq$ actual N	Smaller N_e = faster genetic drift = faster loss of diversity; key parameter for conservation planning
N_e formula (unequal sex ratio)	$N_e = 4N_mN_f / (N_m + N_f)$ where N_m = breeding males; N_f = breeding females	Unequal sex ratio drastically reduces N_e ; e.g. one dominant male = very small N_e
N_e and N relationship	N_e is almost always LESS than actual census N	N_e/N ratio typically 0.1–0.25 in real populations due to unequal reproduction, sex ratio variance
50/500 rule (Franklin 1980)	$N_e \geq 50$ to avoid inbreeding depression short-term; $N_e \geq 500$ to maintain evolutionary potential	Minimum viable population planning guideline; 50 for immediate survival, 500 for long-term evolution

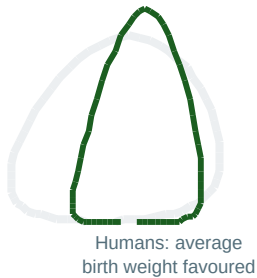
TOPIC 9: EVOLUTION — NATURAL SELECTION, SPECIATION & ADAPTATION

9.1 Types of Natural Selection

THREE TYPES OF NATURAL SELECTION

Stabilising Selection

Intermediate phenotypes favoured; variation REDUCED



Directional Selection

One extreme favoured; population mean SHIFTS



Disruptive Selection

Both extremes favoured; INTERMEDIATES disfavoured

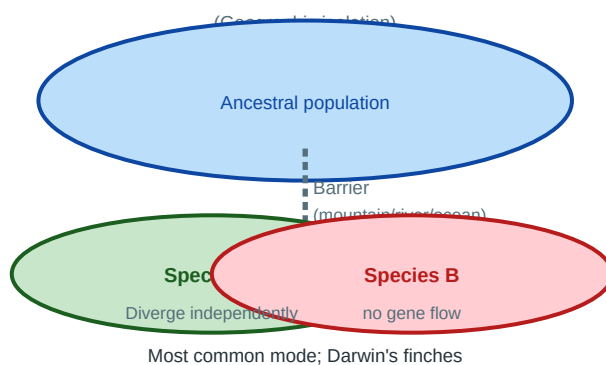


Selection type	Which phenotypes survive	Effect on variation	Population mean	Classic examples
Stabilising selection	INTERMEDIATE phenotypes; extremes eliminated	REDUCES variation (narrower bell curve)	STAYS same (no shift)	Human birth weight (too low or too high = higher mortality); egg size in birds
Directional selection	ONE EXTREME phenotype has highest fitness	Initially reduces then can increase variation	SHIFTS toward favoured extreme	Antibiotic resistance; industrial melanism (peppered moth dark form); horse leg length
Disruptive (diversifying) selection	BOTH EXTREMES survive better than intermediates	INCREASES variation (bimodal distribution)	Can SPLIT into two peaks	Darwin's finches (large + small beaks survive; medium fail); human ABO blood group in some malaria regions
Sexual selection	Traits preferred by opposite sex — even if survival cost	Can increase or decrease variation	Shifts toward sexually selected trait	Peacock tail (female choice); lion mane; stag antlers (male-male competition)
Artificial selection	Humans select for desired traits	Rapid change in selected traits	Rapid shift in direction humans choose	Dog breeds from wolves; crop varieties; livestock breeds

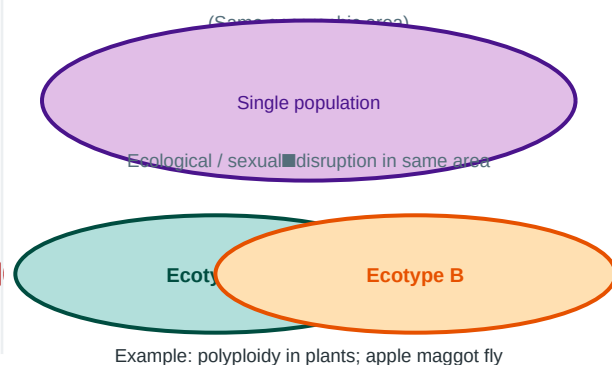
9.2 Speciation

SPECIATION — Allopatric vs Sympatric

ALLOPATRIC SPECIATION



SYMPATRIC SPECIATION



Speciation type	Geographic overlap	Isolation mechanism	Rate	Examples
Allopatric	NO — geographically SEPARATED	Geographic barrier (mountain, river, ocean, desert) prevents gene flow	Slow (thousands of generations)	Darwin's finches (islands); Galapagos iguanas; subspecies becoming species
Sympatric	YES — SAME geographic area	Ecological, behavioural, or chromosomal isolation without physical barrier	Can be fast (polyploidy = instant)	Polyploidy in plants (bread wheat = 6n); apple maggot fly (host race specialisation); cichlid fish in African lakes
Parapatric	Partially — populations ADJACENT	Partial geographic separation; some gene flow at edges; strong selection along gradient	Intermediate	Grass populations on toxic mine tailings adjacent to normal populations
Peripatric	Small peripheral population becomes ISOLATED	Small marginal population isolates; founder effect + selection	Fast due to genetic drift	Island colonisation by birds; endemic species on oceanic islands

9.3 Adaptive Radiation

Adaptive radiation = rapid evolutionary diversification of a single ancestral lineage into many ecological roles (niches), often following entry into new environment or after a mass extinction. Classic examples: Darwin's finches (14 species from one ancestral finch on Galapagos), Hawaiian honeycreepers, cichlid fish in African Great Lakes (500+ species), mammal diversification after dinosaur extinction, Australian marsupials.

9.4 Convergent vs Divergent vs Parallel Evolution

Type	Definition	Produces	Example
Convergent evolution	UNRELATED species evolve SIMILAR traits due to similar selective pressures	Analogous structures (similar function, different origin)	Streamlined body: dolphins (mammal) + sharks (fish); wings: birds+bats+insects; eyes: vertebrates+octopus
Divergent evolution	RELATED species evolve DIFFERENT traits as they adapt to different environments	Homologous structures (same origin, different function)	Vertebrate forelimb: human arm, bat wing, whale flipper, horse leg — same bones, different functions; Darwin's finch beaks
Parallel evolution	RELATED species independently evolve SIMILAR traits in similar environments	Similar traits from shared genetic architecture	Marsupial vs placental equivalents (marsupial mole = placental mole; Tasmanian wolf = grey wolf)

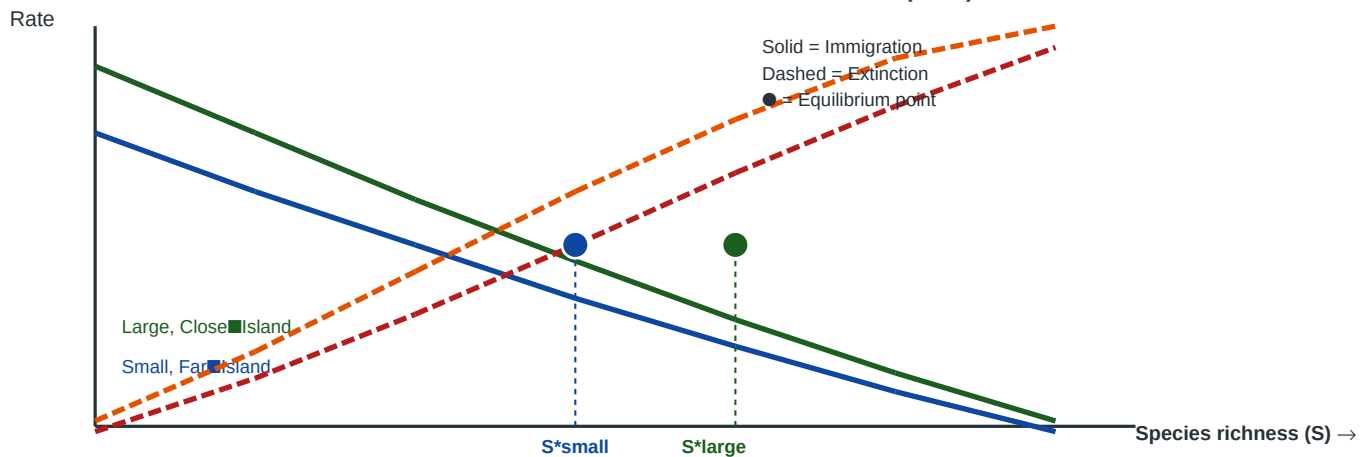
EXAM TIPS — Evolution

1. Stabilising = INTERMEDIATE survives; REDUCES variation; mean STAYS. Most common in stable environments. 2. Directional = ONE EXTREME survives; mean SHIFTS. Example: antibiotic resistance, peppered moth. 3. Disruptive = BOTH EXTREMES survive; variation INCREASES; can lead to speciation. 4. Allopatric (geographic isolation) = MOST COMMON speciation mode. Darwin's finches = classic example. 5. Sympatric speciation via POLYPLOIDY is instantaneous in plants (bread wheat 6n from crossing 3 diploid species). 6. CONVERGENT = unrelated → ANALOGOUS structures. DIVERGENT = related → HOMOLOGOUS structures. 7. Hardy-Weinberg: $q = \sqrt{\text{recessive homozygote frequency}}$. Carrier frequency = $2pq$.

TOPIC 10: APPLIED ECOLOGY — ISLAND BIOGEOGRAPHY, METAPOPULATION & MORE

10.1 Island Biogeography Theory (MacArthur & Wilson, 1967)

ISLAND BIOGEOGRAPHY THEORY — MacArthur & Wilson (1967)



Island Biogeography Theory (MacArthur & Wilson, 1967): The number of species on an island reaches an equilibrium (S^*) determined by the BALANCE between immigration rate (colonisation) and extinction rate. This equilibrium varies with island size and distance from mainland.

Island characteristic	Effect on immigration	Effect on extinction	Effect on S^* (species richness)
Large island	—	LOWER extinction (more habitats, larger populations)	HIGHER S^*
Small island	—	HIGHER extinction (fewer habitats, smaller populations more vulnerable)	LOWER S^*
Close to mainland	HIGHER immigration (shorter distance for colonisers to travel)	—	HIGHER S^*
Far from mainland	LOWER immigration (longer distance, fewer colonisers reach island)	—	LOWER S^*
Large AND close	Highest immigration + lowest extinction	—	HIGHEST S^* (maximum species richness)
Small AND far	Lowest immigration + highest extinction	—	LOWEST S^* (minimum species richness)

Application to conservation: Island biogeography principles apply to habitat FRAGMENTS (forest patches, nature reserves, isolated wetlands) surrounded by an inhospitable 'sea' of human-altered land. Larger, connected reserves = higher species survival. Habitat corridors between reserves simulate island proximity — they reduce extinction by allowing immigration (recolonisation).

SLOSS debate	Definition	Implication
Single Large	One large reserve	Higher diversity per reserve; reduced edge effects; better for wide-ranging species

SLOSS debate	Definition	Implication
Or Several Small	Many smaller reserves	Protects more geographic areas; insurance against catastrophic events; better for rare locally endemic species
Current consensus	Depends on species and context	Connectivity (corridors) between reserves is MORE important than SLOSS argument

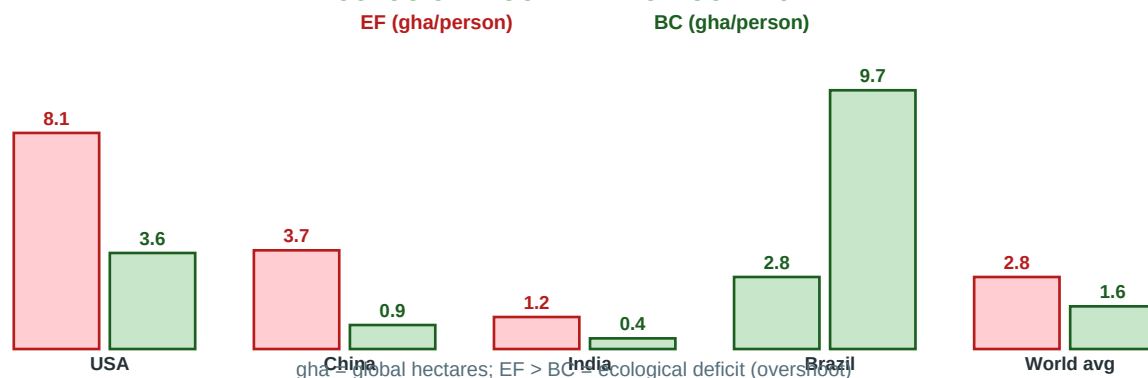
10.2 Metapopulation Theory (Richard Levins, 1969)

Metapopulation = a set of local subpopulations of the same species inhabiting separate habitat patches and connected by dispersal (immigration/emigration). Local subpopulations can go extinct and be recolonised from other patches — 'winking' of local populations. The metapopulation as a whole persists even when individual subpopulations go extinct.

Concept	Definition	Conservation application
Subpopulation	Local breeding group in one habitat patch	Each patch must be large enough to sustain viable local population
Dispersal corridors	Habitat connecting patches allowing movement of individuals	Crucial for recolonisation; without corridors, isolated patches go extinct permanently
Rescue effect	Immigration from nearby patch prevents local extinction	Large source populations rescue small sink populations
Source-sink dynamics	Source: productive habitat (positive growth); Sink: poor habitat (negative growth without immigration)	Protect source habitats first; sink habitats maintained by immigration
Patch occupancy	Fraction of patches occupied at any time	Used to assess metapopulation viability
Critical threshold	Minimum patch occupancy below which metapopulation collapses	Management target: maintain occupancy above threshold

10.3 Ecological Footprint & Biocapacity

ECOLOGICAL FOOTPRINT vs BIOCAPACITY



Ecological Footprint (Mathis Wackernagel & William Rees, 1992): measure of human demand on nature — the biologically productive land and water area needed to produce all resources consumed by a population and to absorb all its waste. Measured in **global hectares (gha)**.

Term	Definition	Global values (approx.)
Ecological footprint (EF)	Total productive area needed to support a population's consumption	World avg: ~2.8 gha/person
Biocapacity (BC)	Total productive area available to generate resources and absorb waste	Earth's BC: ~1.6 gha/person

Term	Definition	Global values (approx.)
Ecological deficit	$EF > BC$ = more consumed than regenerated (overshoot)	World in ecological overshoot since 1970s
Ecological reserve	$BC > EF$ = sustainable use — less consumed than available	Brazil, Russia, Canada have ecological reserves
Earth Overshoot Day	Date when humanity has used all biological resources Earth can regenerate in one year	2023: August 2nd — spending rest of year in deficit

10.4 Restoration Ecology

Concept	Definition	Examples
Ecological restoration	Process of assisting recovery of an ecosystem that has been degraded, damaged, or destroyed	Mine land reclamation, river restoration, mangrove replanting, grassland seeding
Rehabilitation	Partial recovery of ecosystem function without full return to pre-disturbance state	Afforestation of degraded lands; improving water quality of polluted rivers
Reclamation	Making degraded land useful again (not necessarily for wildlife)	Coal mine reclamation for agriculture or urban use
Reference ecosystem	Model for the target state of restoration (historically similar or local undisturbed ecosystem)	Nearby intact forest used as reference for reforestation project
Bioremediation	Using living organisms (bacteria, plants) to detoxify/remove contaminants	Phytoremediation: hyperaccumulator plants remove heavy metals; bacterial degradation of oil spills

10.5 Umbrella, Keystone & Flagship Species

Species type	Definition	Why important for conservation	Indian example
Umbrella species	Species with large habitat requirements; protecting them protects many other species sharing same habitat	Protecting tigers protects vast landscapes and hundreds of other species	Tiger — protects entire forest ecosystem
Keystone species	Species with disproportionately large effect on ecosystem relative to its abundance; ecosystem collapses without it	Identifies critical ecological role; removal causes trophic cascade	Sea otter (Pacific), elephant (African savanna), fig tree (tropical forests)
Flagship species	Charismatic species used as symbol to promote conservation awareness and funding	Public appeal and fundraising; policy influence	Tiger, snow leopard, Indian rhino, great Indian bustard
Indicator species	Species sensitive to environmental change; used to monitor ecosystem health	Early warning of environmental deterioration	Lichens (air quality), freshwater invertebrates (water quality), amphibians (ecosystem health)

EXAM TIPS — Applied Ecology

1. Island biogeography: LARGE + CLOSE = highest species richness. SMALL + FAR = lowest. 2. Equilibrium S^* = where immigration curve INTERSECTS extinction curve. 3. SLOSS: modern consensus = connectivity/corridors MORE important than size debate. 4. Metapopulation: Levins (1969). Source patch = positive growth. Sink = depends on immigration. 5. Ecological footprint unit = global hectares (gha). Earth Overshoot since 1970s. 6. Umbrella (Tiger) $\neq$ Keystone (fig tree). Umbrella = large range. Keystone = disproportionate effect. 7. Bioremediation: bacteria for oil; phytoremediation: hyperaccumulator plants for heavy metals.

UNIT 3 COMPLETE MASTER QUICK REFERENCE — PART 2

Scientists & Key Contributions — Part 2 Topics

Scientist	Year	Contribution	Exam frequency
Norman Myers	1988	Biodiversity hotspot concept; originally 10, now 36 globally	VERY HIGH
E.O. Wilson	1986	Popularised 'biodiversity'; organised first BioDiversity Forum; HIPPO framework; island biogeography	VERY HIGH
R.H. MacArthur & E.O. Wilson	1967	Island Biogeography Theory: $S^* = \text{balance of immigration and extinction}$	VERY HIGH
Richard Levins	1969	Metapopulation theory; patch dynamics	HIGH
Godfrey Hardy & Wilhelm Weinberg	1908	Hardy-Weinberg equilibrium: $p^2 + 2pq + q^2 = 1$	VERY HIGH
Mathis Wackernagel & William Rees	1992	Ecological footprint concept; global hectares	HIGH
Charles Darwin	1859	Natural selection; On the Origin of Species; Galapagos finch observations	HIGH
R.H. Whittaker	1960	Alpha, beta, gamma diversity — coined all three terms	VERY HIGH
Franklin (Soule & Simmerloff)	1980	50/500 rule for minimum viable population genetics	MEDIUM
Connell & Slatyer	1977	Three models of succession: facilitation, tolerance, inhibition	HIGH
Pielou	1966	Evenness index $J = H'/\ln(S)$	MEDIUM
C.S. Holling	1959	Functional response of predators; resilience concept	MEDIUM

Key Data — Must Memorise for Exam

Fact	Value/Detail	Exam frequency
Global biodiversity hotspots	36 hotspots (Conservation International, 2023)	VERY HIGH
Hotspot criteria — endemic plants	$\geq 1,500$ endemic vascular plant species	VERY HIGH
Hotspot criteria — habitat loss	$\geq 70\%$ of original native habitat lost	VERY HIGH
India's biodiversity hotspots	4: Western Ghats+SL, Eastern Himalayas, Indo-Burma, Sundaland	VERY HIGH
Global hotspots: % of Earth's land	$\sim 2.5\%$ of land surface	HIGH
Global hotspots: % endemic plants	$> 50\%$ of plant endemics	HIGH
Project Tiger launch	1 April 1973; 54 reserves (2023)	VERY HIGH
Wildlife Protection Act	1972; Schedule I = highest protection	VERY HIGH
Biological Diversity Act	2002; NBA = National Biodiversity Authority, Chennai	HIGH
Chilika Lake	India's first Ramsar site (1981), Odisha	HIGH
India's Ramsar sites	75+ (increasing annually)	HIGH

Fact	Value/Detail	Exam frequency
India's biosphere reserves	18 total; 12 UNESCO World Network	HIGH
India's tiger reserves	54 (as of 2023)	MEDIUM
India's national parks	106 (approximately)	MEDIUM
India's wildlife sanctuaries	567 (approximately)	MEDIUM
H-W equation	$p + q = 1$; $p^2 + 2pq + q^2 = 1$	VERY HIGH
Finding q from phenotype	$q = \sqrt{\text{(recessive homozygote frequency)}}$	VERY HIGH
Carrier frequency	$2pq$	VERY HIGH
Allopatric speciation example	Darwin's 14 Galapagos finch species	HIGH
Convergent evolution example	Dolphin (mammal) + shark (fish) body shape — analogous	HIGH
Divergent evolution structure	Vertebrate forelimb — homologous	HIGH
Ecological footprint unit	Global hectares (gha)	HIGH
World average ecological footprint	~2.8 gha/person	MEDIUM
Earth's biocapacity	~1.6 gha/person	MEDIUM
Earth overshoot started	1970s (humanity using >1 Earth since then)	MEDIUM
50/500 rule	$N_e \geq 50$ (short-term); $N_e \geq 500$ (long-term evolution)	MEDIUM

Complete Mnemonics — Part 2

Topic	Mnemonic / Memory Aid
India's 4 hotspots	WISH = Western Ghats, Indo-Burma, Sundaland (Andaman), Himalayas (Eastern)
HIPPO threats	H-Habitat loss (worst), I-Invasive species, P-Pollution, P-Population (human), O-Overexploitation
IUCN order (most to least threatened)	EX-EW-CR-EN-VU-NT-LC = 'Every Elephant Can Eat Very Nice Leaves' (+ DD + NE)
THREATENED = 3 categories	CR + EN + VU = THREATENED. Remember: 'CEV = Critical, Endangered, Vulnerable'
Biosphere reserve zones	C-B-T from inside out: Core (no humans), Buffer (research), Transition (people live)
Hardy-Weinberg find q	$q = \text{SQUARE ROOT of the recessive visible phenotype frequency}$ (q^2 is the affected group)
Carrier frequency	$2pq$ = carrier. Two parents, p and q, making carrier = $2pq$. Always the LARGEST genotype group when alleles are common.
Convergent vs divergent	CONVERGENT = looks Similar but UNrelated (ANALOGOUS structures). DIVERGENT = related but look Different (HOMOLOGOUS).
Allopatric vs sympatric	ALLO = Apart (geographic separation). SYMPATRIC = Same place (reproductive isolation within same area).
Island species richness	LACC: Large+Close = most species. Small+Far = fewest species.
Keystone vs umbrella species	KEYSTONE = disproportionate EFFECT (remove it = collapse). UMBRELLA = large RANGE (protect it = protect many).
Natural selection types	SDD: Stabilising (narrow), Directional (shift), Disruptive (split). Mean: stays, shifts, splits.
Ecological footprint units	gha = Global Hectares. 'Going Hungry Anyway' = using more than Earth provides (overshoot)

Complete Unit 3 Master Summary Tree

UNIT 3: ENVIRONMENTAL BIOLOGY — COMPLETE MAP

■■■ PART 1 (Topics 1-5):

- 1. ECOLOGY BASICS: Haeckel(1866), Tansley(1935), autecology vs synecology
- 2. AUTECOLOGY: Liebig(Minimum/scarcest), Shelford(Tolerance/spectrum); stenothermal/eurythermal
- 3. POPULATION ECOLOGY: $dN/dt=rN(J\text{-curve})$ vs $dN/dt=rN(K-N)/K(S\text{-curve})$; max growth= $K/2=MSY$
- Age pyramids: triangular(growing), column(stable), inverted(declining)
- Survivorship: Type I(humans-late loss), Type II(birds-constant), Type III(fish-early loss)
- r-strategy(many/small/fast) vs K-strategy(few/large/slow); Allee effect; MVP; metapopulation
- 4. COMMUNITY ECOLOGY:
Interactions(++mutualism,+0commensal,-0amensalism,+predation/parasitism,--competition)
- Gause's Competitive Exclusion (Paramecium) → niche partitioning; character displacement
- Batesian mimicry(harmless mimics harmful) vs Müllerian(both harmful mimic each other)
- Lotka-Volterra oscillation; predator LAGS prey; arms race coevolution; parasitoids
- 5. SUCCESSION: Primary(bare rock, lichens, slow) vs Secondary(soil, weeds, fast)
- Hydrosere: Phytoplankt→Submerged→Floating→Reed→Marsh→Shrub→Forest
- Psammosere: Marram grass pioneer. Halosere: Salicornia pioneer.
- Trends: biomass↑, diversity↑, NPP peaks then↓, P:R→1, complexity↑

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■■■ PART 2 (Topics 6-10):

- 6. BIODIVERSITY: α (within site), β (between sites= γ/α), γ (landscape total)
- $H'=-\sum(\pi \times \ln \pi)$; Simpson $D=\sum \pi^2$; Evenness $J=H'/\ln(S)$
- Hotspots: 36 globally; ≥ 1500 endemic plants AND $\geq 70\%$ habitat loss (Myers,1988)
- India 4 hotspots: WISH = Western Ghats, Indo-Burma, Sundaland, Himalayas
- HIPPO: Habitat(worst), Invasive, Pollution, Population, Overexploitation
- 7. CONSERVATION: In-situ(NP, Sanctuary, BR) vs Ex-situ(zoo, seed bank, captive breeding)
- IUCN: EX→EW→CR→EN→VU→NT→LC; THREATENED=CR+EN+VU
- BR zones: Core(strict)→Buffer(research)→Transition(people); 18 India, 12 UNESCO
- Laws: WPA1972(Sch.I), FCA1980, BDA2002(NBA-Chennai), EPA1986
- Conventions: CITES1975(Appl=no trade), Ramsar1971(wetlands, Chilika=1st India), CBD1992
- 8. GENETICS: $H-W(p^2+2pq+q^2=1)$; $q=\sqrt{q^2 \text{ from phenotype}}$; carriers=2pq
- Drift: Founder effect, Bottleneck effect; Inbreeding depression; N_e
- 9. EVOLUTION: Stabilising(narrow), Directional(shift), Disruptive(split)
- Allopatric(geographic isolation, commonest) vs Sympatric(polyploidy, same area)
- Convergent=analogous=unrelated. Divergent=homologous=related. Adaptive radiation.
- 10. APPLIED: Island biogeography(Large+Close=most S^* ; MacArthur+Wilson,1967)
- Metapopulation(Levins,1969); source-sink; corridors critical
- Ecological footprint(gha, Wackernagel&Rees,1992); Earth in overshoot since 1970s
- Umbrella(tiger=large range) vs Keystone(fig tree=disproportionate effect) vs Flagship

YOU HAVE COMPLETED UNIT 3 — TEST YOURSELF

Before your exam, you must be able to: ✓ Name all 4 India hotspots and their key features without hesitation ✓ Recall all 9 IUCN categories in order and which 3 = 'threatened' ✓ Calculate q and 2pq from a phenotype frequency using Hardy-Weinberg ✓ Draw the biosphere reserve three-zone diagram from memory ✓ Distinguish allopatric from sympatric speciation with examples ✓ Explain why S^* is highest on large, close islands ✓ State the 50/500 rule and what N_e means ✓ Name the Project Tiger launch year (1973) and number of reserves (54) ✓ State the Wildlife Protection Act year (1972) and highest schedule (I) ✓ List HIPPO factors in order of severity with Habitat loss as greatest NEXT RECOMMENDED: Unit 5 (Pollution & Control) — most PYQs of any unit | Unit 8 (Laws & Policies) — direct factual questions, high ROI | Unit 2 (Environmental Chemistry) — BOD/COD numericals